Run 224489: Cross-Wiring Fix Confirmed, First Plastic-MIP Calibration from Liquid-Scintillator Triples, and the Second Plastic HV Scan

n_TOF EAR2 X17 campaign, run 224489 (July 17, 2026)

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analysis and document prepared with Claude (Anthropic AI assistant)

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Abstract

Run 224489 (1750 bunches, EAR2) is the first run with the liquid scintillators (LIQ) in the readout, the first after the A/D cabling fix, and the first with the plastics on a linear fan-in/fan-out (FIFO) instead of the BNC-T split; it also repeats the nine-step plastic HV ladder of run 224466. Five headline results. (1) The cross-wiring fix is confirmed: the ± 4 ns equal-amplitude duplication band on arms A/D is gone (flagged fractions $\leq 4\%$, the B-arm control level, vs. 23–31% before), all four arms now show equal wall–plastic coincidence excess, and A/D display clean MIP bumps with no veto — the standing A/D deficit was the cabling. (2) Triple coincidences WAL$\times$PSS$\times$LIQ select through-going particles and yield the *first plastic-MIP calibration*: MIP (linear-space MPV) amplitudes at nominal HV span 3.3 mV (DR) to 10.3 mV (AL), i.e. 0.65–2.05 mV/MeV against the expected 5.05 MeV deposit in 2.5 cm of PVT; the DR and BL MIPs sit below the recommended 4.9 mV trigger threshold. The extraction is validated against the no-MIP-selection coincident median: both follow the same gain power law, the MIP sitting a factor $\sim$10 below. (3) The FIFO recovers only $\times 1.40$ (fleet geometric mean; 1.13–1.65 per PMT) rather than the naive $\times 2$. (4) The liquids are timed in: every arm shows a sharp LIQ coincidence peak, the LIQ pulse leading the plastic by $\sim$30 ns; and the suspected LIQ gain gradient is real — a 1.5–2$\times$ amplitude variation along the vessel axis, higher near the PMT, whose direction flips with the surveyed vessel orientation. (5) All 64 wall–plastic time offsets shifted by a single common -22.27 ns (rms 0.22 ns), while the -60 ns satellite bump kept its *absolute* position and grew to 19–34% of the main peak: it is not a plastic-cable reflection and warrants a scope check at the FIFO. A new HV-equalization table supersedes the run-224466 one.

1 Run overview and hardware changes

Run 224489 was taken 2026-07-17 (17:39–19:17 local) during the second plastic HV scan (`plastic_scan_2`): the same nine-step ladder as run 224466 — pass 1 at 1600$\to$1200 V in 100 V steps, pass 2 at 1550$\to$1250 V — merged into a uniform 50 V grid, followed by a $\sim$12 min post-scan window at the per-channel nominal HV (AL 1325, AR 1275, BL 1325, rest 1300 V). Four things changed since 224466:

1. **Cabling fix**: the crossed/duplicated same-side channels on arms A and D (identified after runs 224404/224460) were re-cabled.
2. **FIFO**: the plastic signals now reach DAQ and trigger through a linear fan-in/fan-out instead of a passive BNC-T split, adding $\sim$16 ns of cable and (naively) a factor 2 in amplitude.
3. **Liquid scintillators**: trees LIQA–D (one channel each) appear for the first time, with 5.3 M (C) to 21.6 M (B) hits.

4. **SiPM flash gating ON for the whole run** (no mid-run boundary, unlike 224466).

Processing used R. Mucciola's new PSA (`UserInput_2026_EAR2_X17.h`, pulse-shape fitting for WALL and LIQ). Data quality is excellent: no SiPM-wall outage (0 of 1750 bunches, vs. 46% lost in 224404), γ-flash at $11.62\,\mu$s on all arms, and all 44 channels alive. All results below use beam-on, wall-active bunches and, for spectra, hits more than 0.1 ms after the γ-flash. Bunches are assigned to HV steps by matching PKUP wall-clock times against the CAEN HV log (Fig. 1).

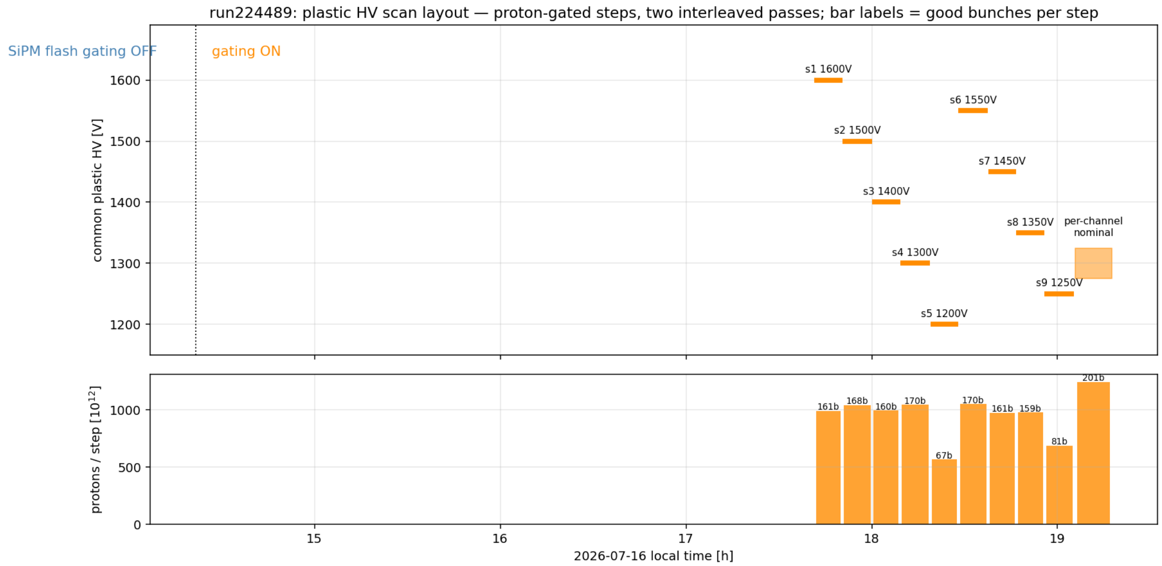

Figure 1: HV staircase vs. wall clock with delivered protons per step. The run has no usable pre-scan window (the first raw file closed one minute before the scan started); the post-scan *end_nominal* window (19:05–19:17) provides the operating-point reference.

2 The cross-wiring fix is confirmed; the A/D question is closed

In runs 224404/224460, arms A and D showed 6–8× fewer true wall–plastic coincidences than B/C and no MIP bump, and a duplication veto study found 23–31% of hits on the affected channels accompanied by an equal-amplitude partner one group over within ± 4 ns (Table 1). In run 224489 every one of those signatures is gone:

flagged fraction (signal pairs)	ch1	ch2	ch3	ch4	ch5	ch6	ch7	ch8
arm A, run224460	0.012	0.029	0.029	0.047	**0.306**	0.049	**0.325**	0.026
arm A, run224489	0.011	0.026	0.026	0.042	0.036	0.044	0.024	0.023
arm D, run224460	0.027	**0.311**	0.043	**0.265**	**0.308**	0.043	**0.343**	0.021
arm D, run224489	0.024	0.021	0.039	0.039	0.038	0.038	0.025	0.019
arm B (control), run224489	0.011	0.008	0.016	0.011	0.029	0.024	0.026	0.024

Table 1: Duplication-veto flagged fractions. The 23–34% bands on the crossed A/D channels (bold) collapse to the B-arm control level after the fix.

Simultaneously, the wall–plastic excess is now equal on all four arms (1.79/1.71/1.81/1.75 M on A/B/C/D within ± 10 ns) and the A/D coincident wall spectra show clean MIP peaks with *no* veto applied (Fig. 2). The conclusion is loud and simple: **the A/D coincidence deficit was the cabling, and it is fixed.** The wall-MIP calibration now covers all four arms.

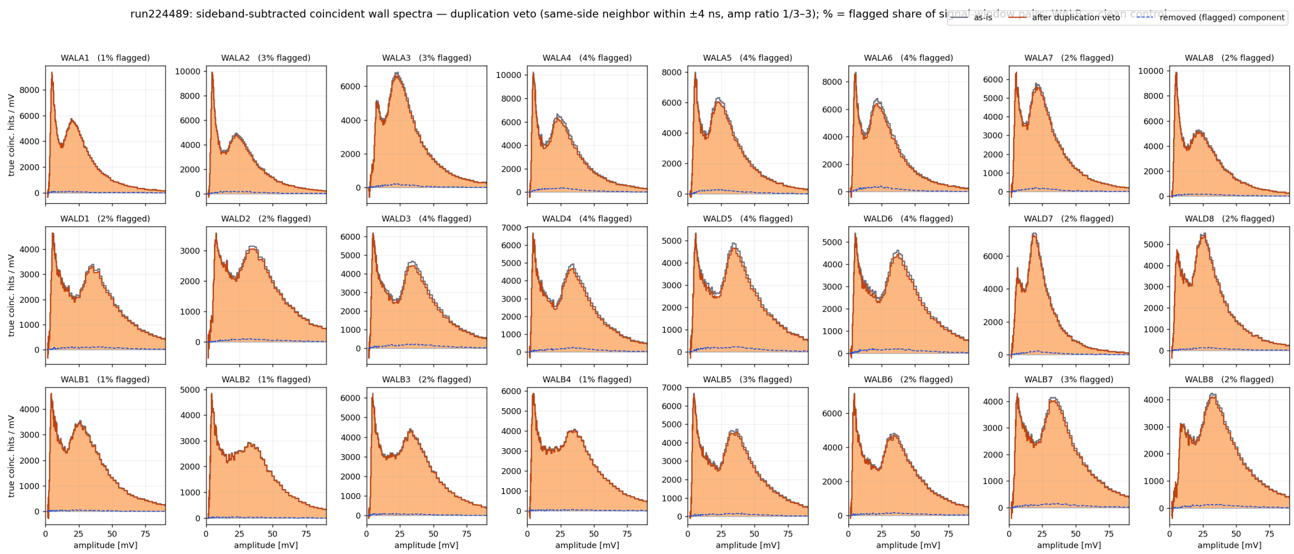

Figure 2: Run 224489 duplication-veto autopsy: sideband-subtracted coincident wall spectra per channel, as-is (colored) vs. veto survivors (black dashed). Unlike 224460, the two are nearly identical everywhere and every channel shows a MIP bump.

3 Plastic HV scan: gain laws, FIFO ratio, new equalization

3.1 Wall MIP is again immune to the plastic HV

The channel-summed, sideband-subtracted wall MIP peak moves by at most one histogram bin ($\pm 2.6\%$) on every arm across the full 1200–1600 V plastic range (Fig. 3) — reproducing the 224466 result, now including arm A. The wall energy scale is independent of the plastic operating point.

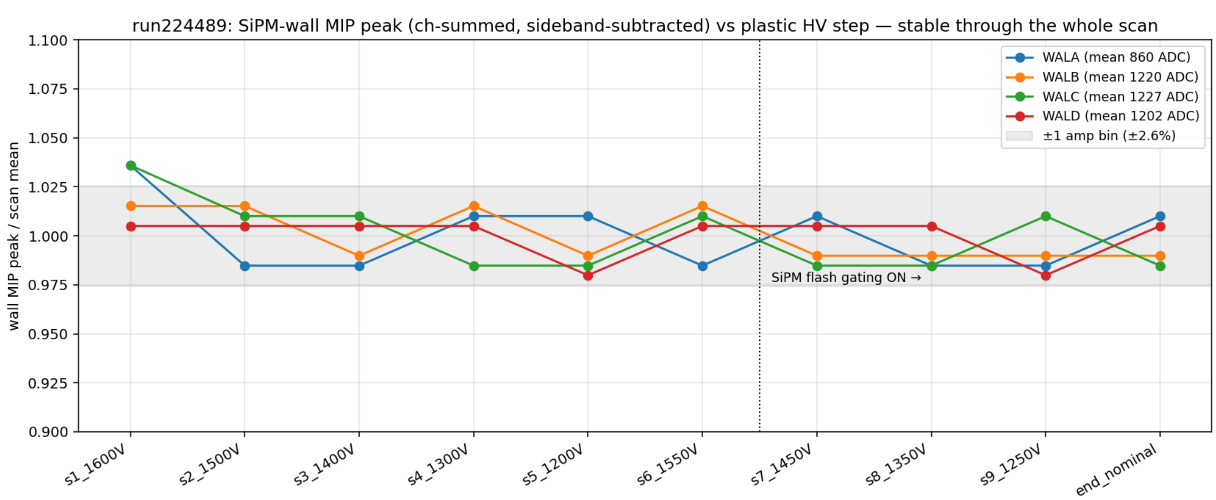

Figure 3: Wall MIP peak position vs. plastic HV step, per arm.

3.2 Per-PMT gain power laws

Wall-tagged, sideband-subtracted plastic medians follow clean power laws $G \propto V^n$ with $n = 3.8$–7.1 (Fig. 4), passes consistent. Two cautionary tales repeat: the *inclusive* medians of PSSB1 and PSSD2 are non-monotonic in V (apparent $n < 1$) — pure threshold bias — while their coincident fits are perfectly ordinary ($n = 5.2$, 6.4). Inclusive medians are not a gain proxy.

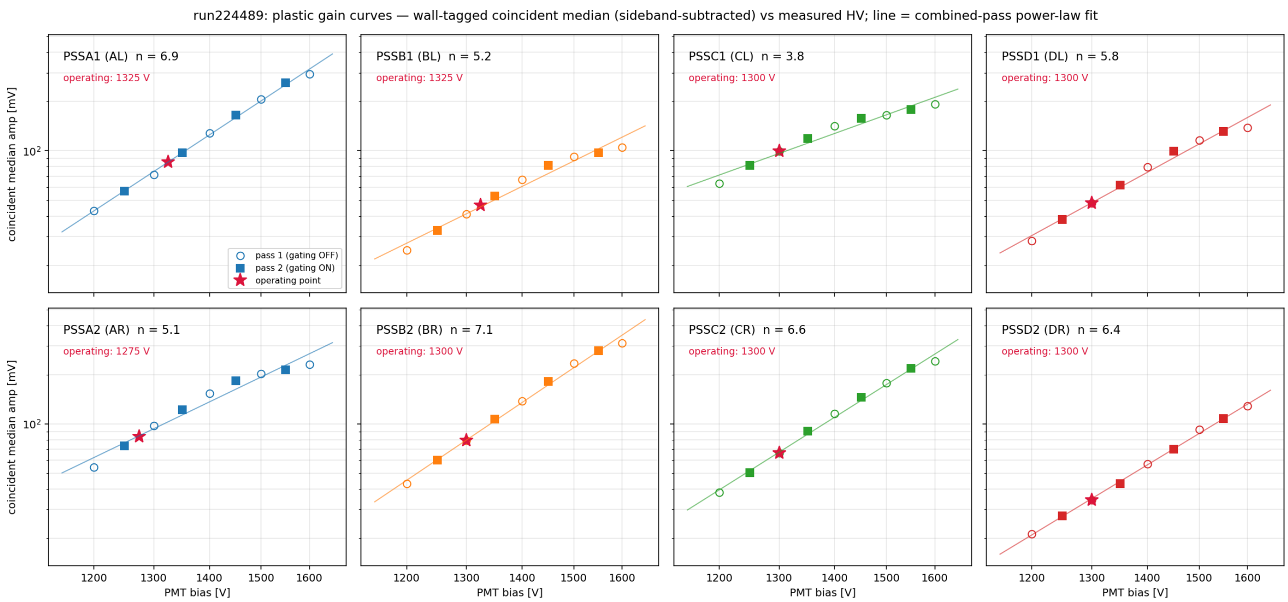

Figure 4: Coincident-median gain curves per PMT with power-law fits; the red star marks the operating point in the post-scan window.

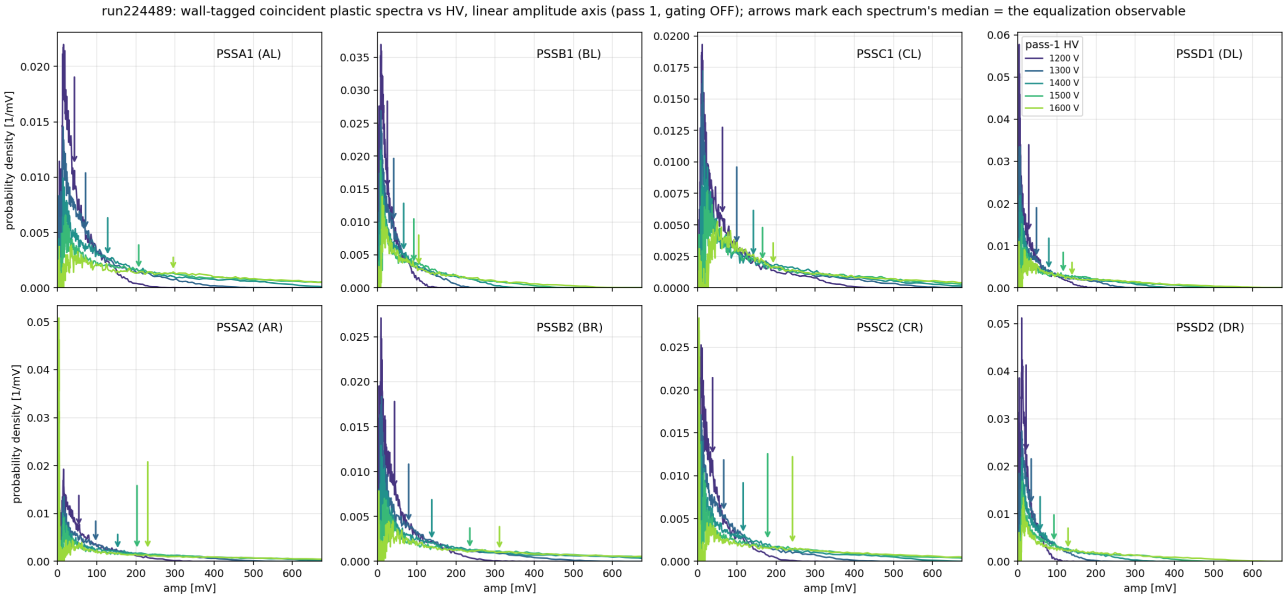

Figure 5: Wall-tagged plastic spectra per step (linear mV axis): the spectral shape marches coherently with HV.

3.3 The FIFO gain ratio is 1.4, not 2

Comparing coincident medians at equal HV between 224489 (FIFO) and 224466 (BNC-T split) measures the signal-path change directly (Fig. 6). The fleet geometric mean is ×1.40, but the per-PMT values span 1.13 (DL) to 1.65 (AL) and are roughly flat in V. The naive expectation — removing a passive T halves the signal, so the FIFO should double it — is wrong in detail; the FIFO's per-channel drive and the old T's actual splitting ratio evidently differ channel by channel. Any gain bookkeeping must use the measured per-PMT ratios, not a global factor 2.

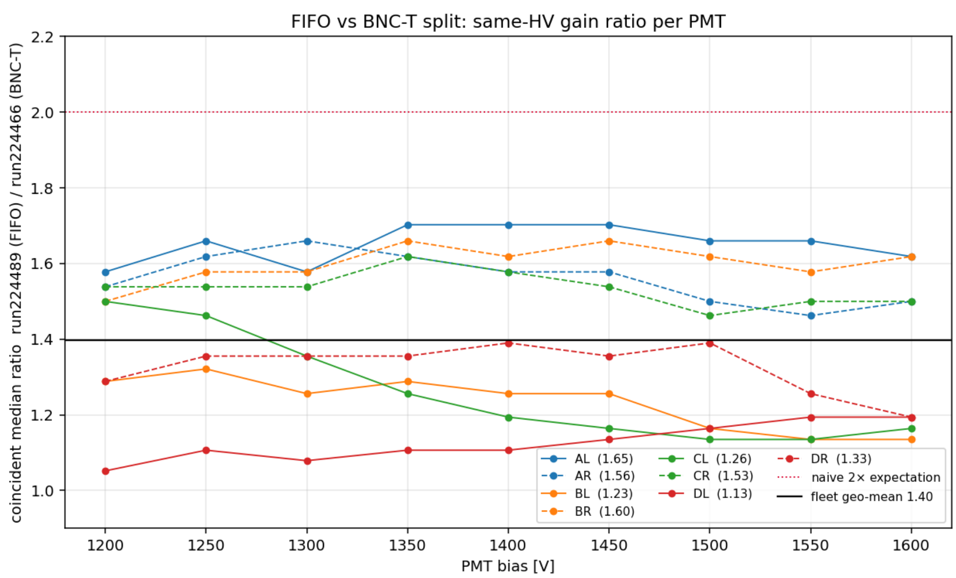

Figure 6: Same-HV coincident-median ratio, run224489/run224466, per PMT.

3.4 New HV equalization and trigger thresholds

Equalizing to the fleet geometric-mean coincident median (64.2 mV) with each PMT's own n gives Table 2; it supersedes the 224466 table (the ordering is unchanged: CL hottest, DR coldest, spread 2.9×). At the equalized gains the common trigger threshold keeping 99% of the wall-tagged population on every PMT is 4.9 mV (95%: 7.1 mV) (Fig. 7). Exported to the DAQ repo (`calibrations/pss/`, `pss_trigger/`).

PMT	V_{now} [V]	median [mV]	n	V_{sugg} [V]	ΔV [V]
PSSA1 (AL)	1325	85.4	6.94	1271	−54
PSSA2 (AR)	1275	83.6	5.08	1211	−64
PSSB1 (BL)	1325	46.7	5.19	1409	+84
PSSB2 (BR)	1300	79.2	7.10	1262	−38
PSSC1 (CL)	1300	99.8	3.80	1158	−142
PSSC2 (CR)	1300	66.5	6.64	1293	−7
PSSD1 (DL)	1300	47.8	5.77	1368	+68
PSSD2 (DR)	1300	34.4	6.39	1433	+133

Table 2: HV equalization to the fleet geo-mean coincident median (2089 ADC = 64.2 mV), run 224489. Supersedes the 224466 table.

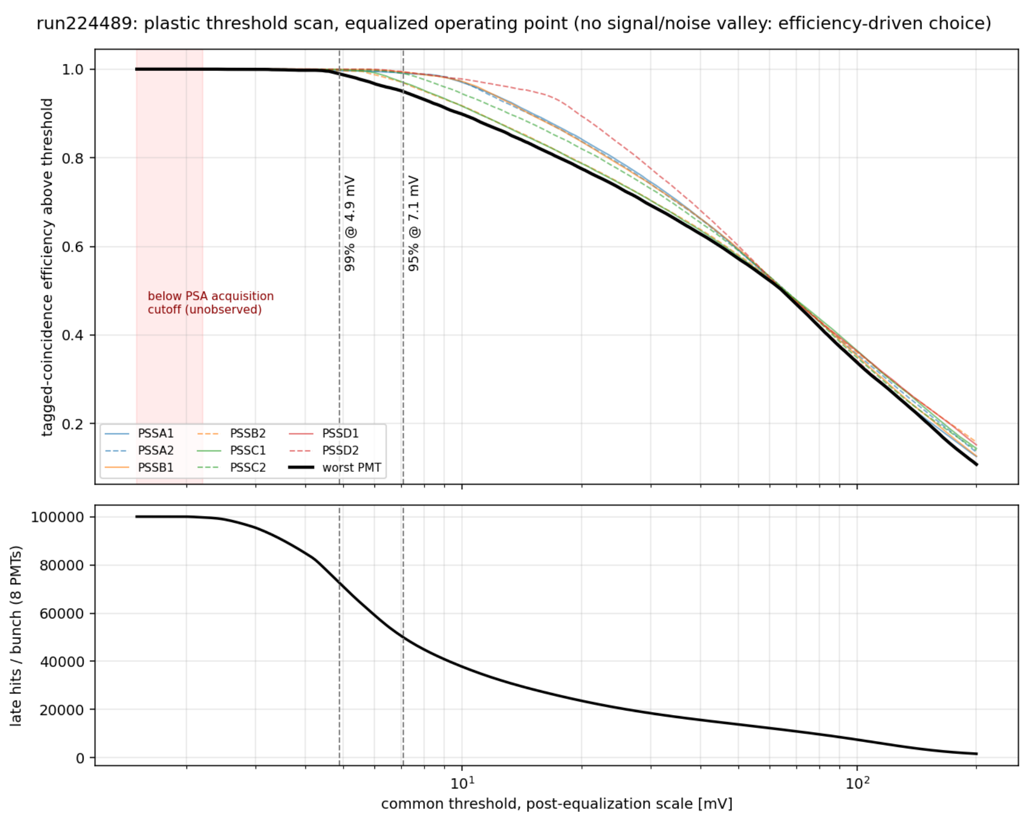

Figure 7: Common-threshold scan at equalized gains.

4 The liquids: timed in on every arm

The LIQ channels had never been timed. A wide (±400 ns) coincidence scan against the same-arm wall and plastic finds a sharp peak on every arm (Fig. 8): the LIQ pulse *leads* the plastic by 29.5–32.8 ns and sits within ±3.3 ns of the wall. Per-channel offsets (8 wall channels × arm, plus both plastic bars; rms 3.2–4.5 ns) are written to `calib/liq_offsets_run224489.json` — a new file, so no existing consumer of the wall–plastic offsets breaks (Fig. 9).

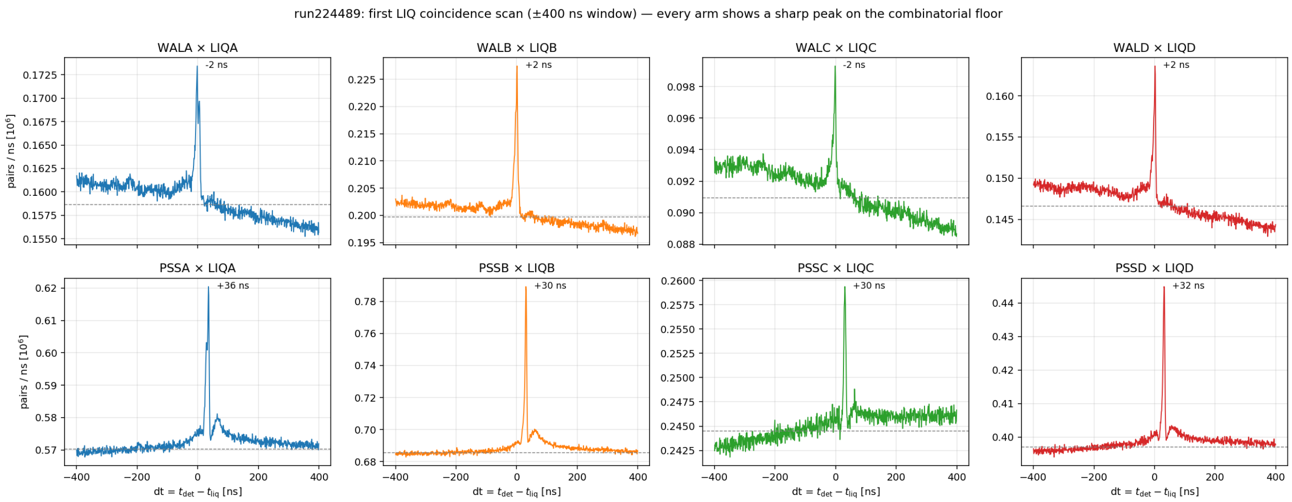

Figure 8: First LIQ coincidence scan, ±400 ns: WAL×LIQ (top) and PSS×LIQ (bottom) per arm.

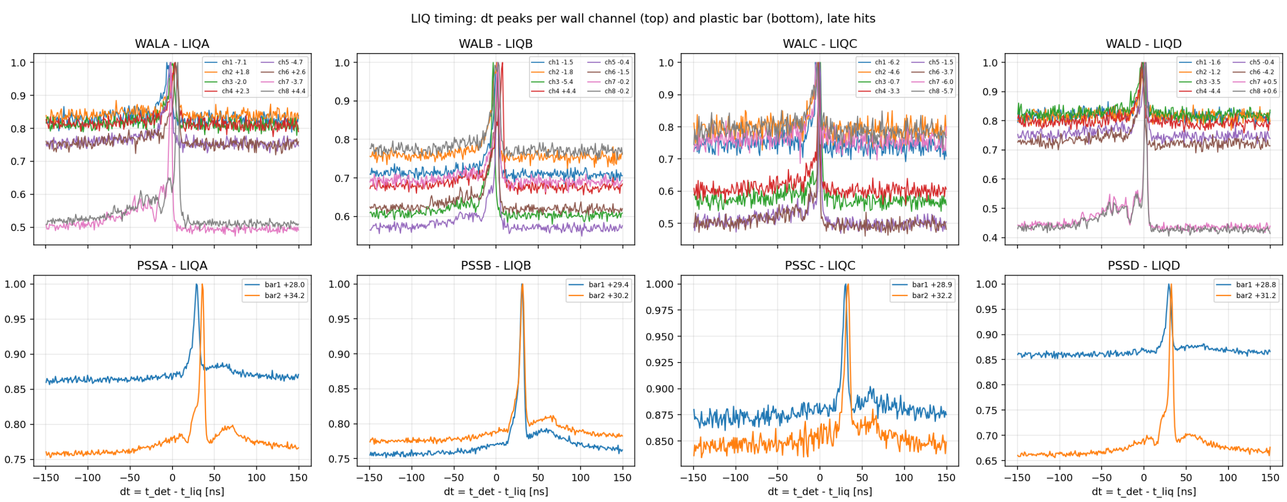

Figure 9: LIQ timing peaks per wall channel (top row) and per plastic bar (bottom row).

5 Triple coincidences: the first plastic MIP

A wall–plastic coincidence MIP-tags only the *wall* (the plastic is behind the wall, so its spectrum in pairs is just a hit spectrum). With the liquids live, a third tag behind the plastic closes the telescope:

the as-built stack is MM $\to$ SiPM wall $\to$ plastics $\to$ single LS layer, so a WAL$\times$PSS$\times$LIQ triple selects through-going particles and the *plastic* spectrum in triples is a genuine MIP-candidate spectrum (Fig. 10).

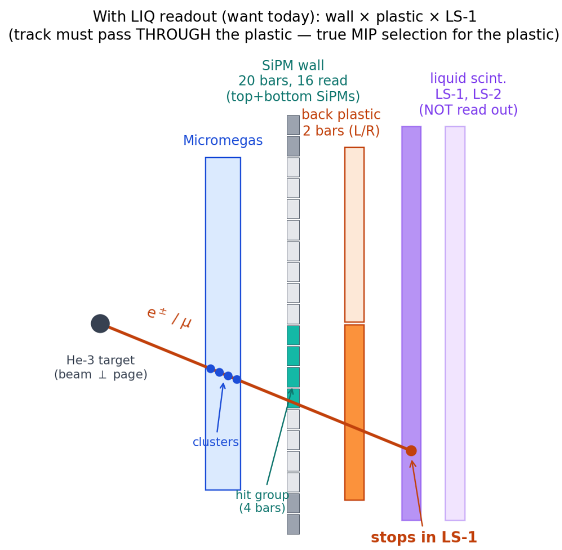

Figure 10: The triple-coincidence telescope.

Triples are built from wall–plastic pairs ($|dt_{\mathrm{cal}}| \leq 8$ ns after the new offsets; sideband +20..+120 ns) matched to LIQ hits near the plastic time ($|dt_{\mathrm{LIQ}}| \leq 10$ ns around the per-arm peak; double sideband 25–125 ns both sides), and every spectrum is double sideband-subtracted in both dt's. Net triple counts per PMT range from 9.2 k (CL) to 49.5 k (BR) — but split over the HV steps this is only $\sim$2–6 k per PMT per step, so the individual per-step spectra (Fig. 12) are statistically noisy; they show the spectral hump marching with HV but are not, by themselves, compelling.

The compelling test is gain alignment (Fig. 11): scaling each step's spectrum to its nominal-voltage equivalent (a multiplication by $(V_{\mathrm{nom}}/V)^n$, i.e. a pure translation in log amplitude) must make a *physical* peak coincide across steps, while the fixed-ADC threshold edge moves from step to step. That is exactly what the data do: the aligned steps pile up at a common position and their sum — one high-statistics spectrum per PMT — shows a single dominant peak at the adopted MIP value, 3–6$\times$ above even the worst step's aligned threshold edge. A threshold artifact cannot produce this behavior. (The aligned stacks also reveal a genuine secondary component at $\sim$100–300 mV equivalent, most visible on AR/BR/DR — large energy deposits, composition not yet identified.) As a further systematic check, adding the LIQ tag leaves the *wall* spectrum in triples unchanged (Fig. 13): the third tag purifies without biasing.

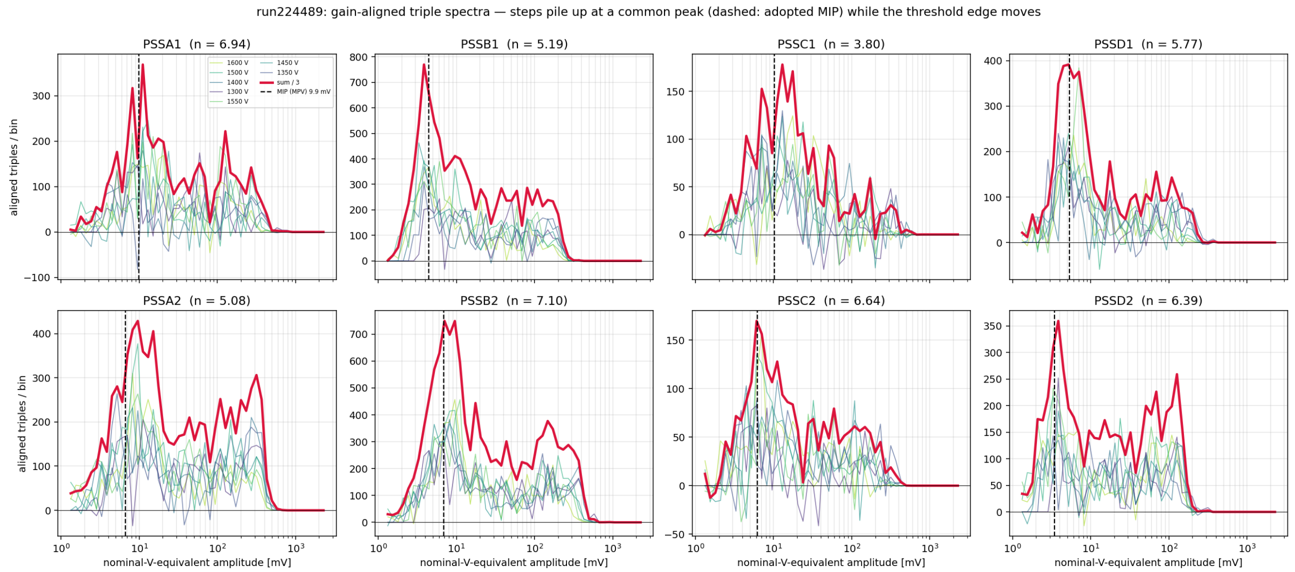

Figure 11: Gain-aligned triple-tagged plastic spectra: thin curves are individual steps ($V \geq 1300\,\mathrm{V}$) scaled to nominal-V equivalent, the red curve is their sum (/3), the dashed line the adopted MIP. A physical peak aligns; the threshold edge does not.

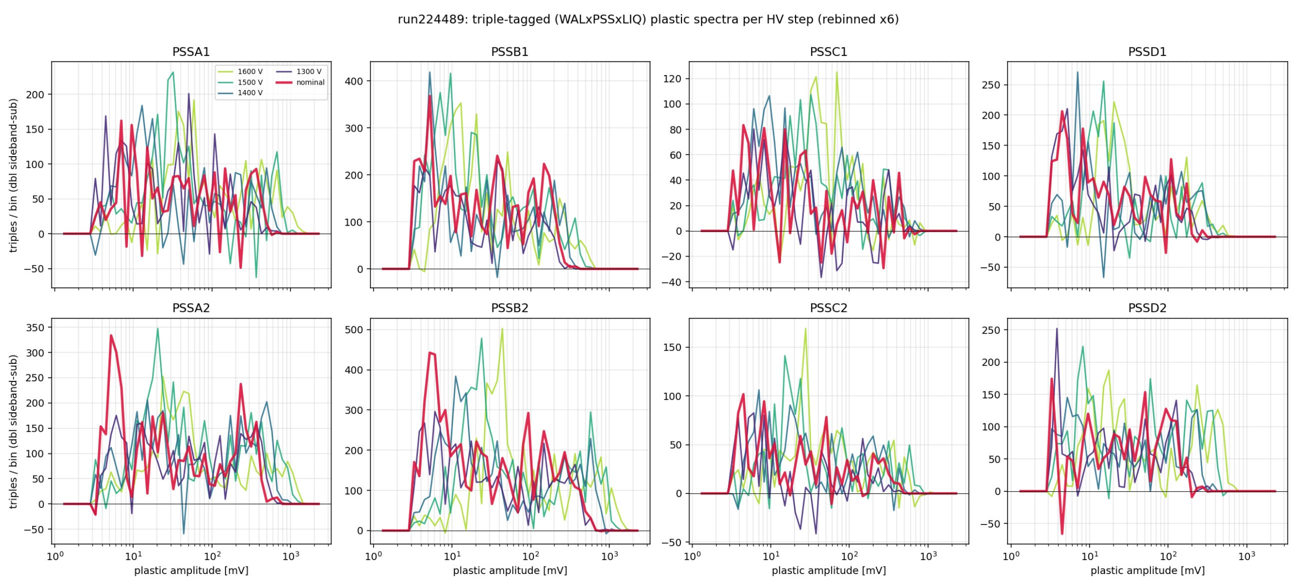

Figure 12: Triple-tagged plastic spectra per HV step (double sideband-subtracted, rebinned ×6; readable subset of steps shown). The MIP hump moves coherently with HV; the red curve is the nominal-HV window.

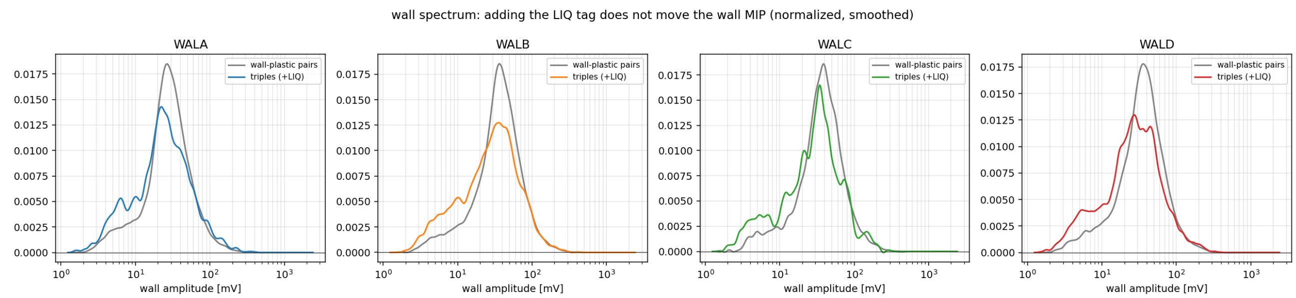

Figure 13: Wall spectra in wall–plastic pairs vs. triples (normalized): the LIQ tag does not move the wall MIP.

The clearest view is one HV step per panel on a *linear* amplitude axis with coarse bins and propagated Poisson errors (Figs. 14–15; all eight PMTs in `figures/19_triples/steps_linear/`): a single peak, significant against its error bars, tracking the power-law expectation panel after panel while the fixed acquisition threshold (gray) stays put. These linear spectra also exposed a convention bias in the first-pass numbers: the mode of a log-binned spectrum ($dN/d\log A = A\,dN/dA$) sits systematically 20–40% above the true linear-space MPV of dN/dA, which is the quantity a Landau MPV convention means. The adopted calibration is therefore the peak of the *gain-aligned summed* linear spectrum (every $V \geq 1400\,\mathrm{V}$ step scaled to its nominal-V equivalent and summed — one $\sim$15 k-triple spectrum per PMT), located with a lightly smoothed peak finder; the statistical error is the standard deviation of the peak over Poisson bootstraps of the four sideband components, and the per-step scatter (Table 3 "spread") measures transport consistency. At these statistics different reasonable peak estimators genuinely differ by 10–30% on the weaker PMTs — that spread, not the bootstrap error alone, is the honest uncertainty of this first calibration.

The decisive trust check is Fig. 16: the per-step MIP MPVs plotted against the *no-MIP-selection* pair-coincident median (the $\sim$100$\times$-statistics gain proxy from the HV scan). Both follow the same power law in V — parallel lines in log-log, n_{MIP} consistent with n_{median} within the low-stats fit errors — with the MIP sitting a factor 8–15 below the median (physical: the pair-tagged plastic spectrum is dominated by larger deposits than a through-going MIP). A noise artifact would not scale with HV like a gain.

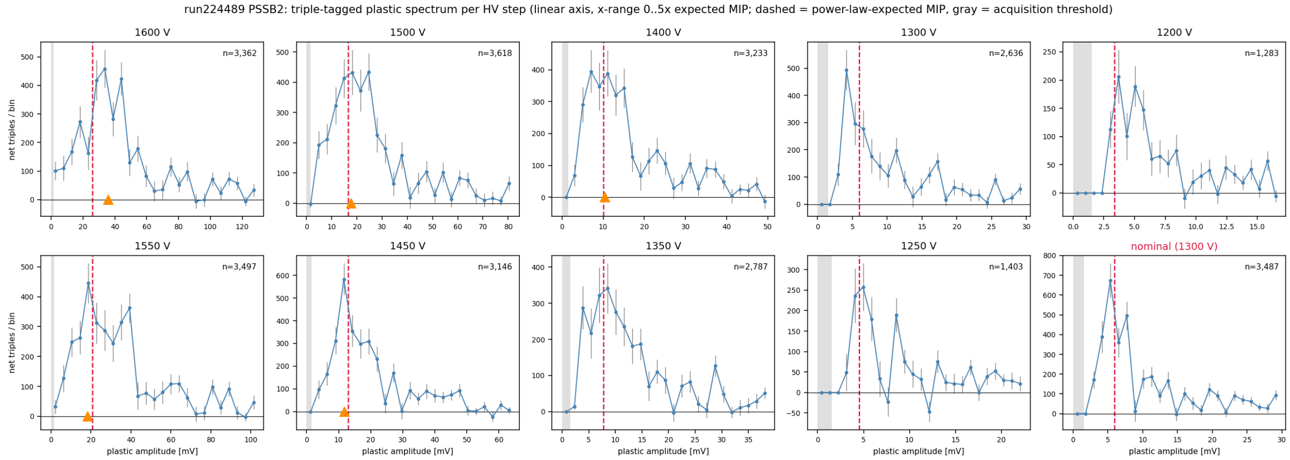

Figure 14: PSSB2: triple-tagged plastic spectrum, one HV step per panel, linear amplitude axis, propagated errors. Dashed: power-law-expected MPV; orange triangle: per-step fitted MPV; gray band: acquisition threshold.

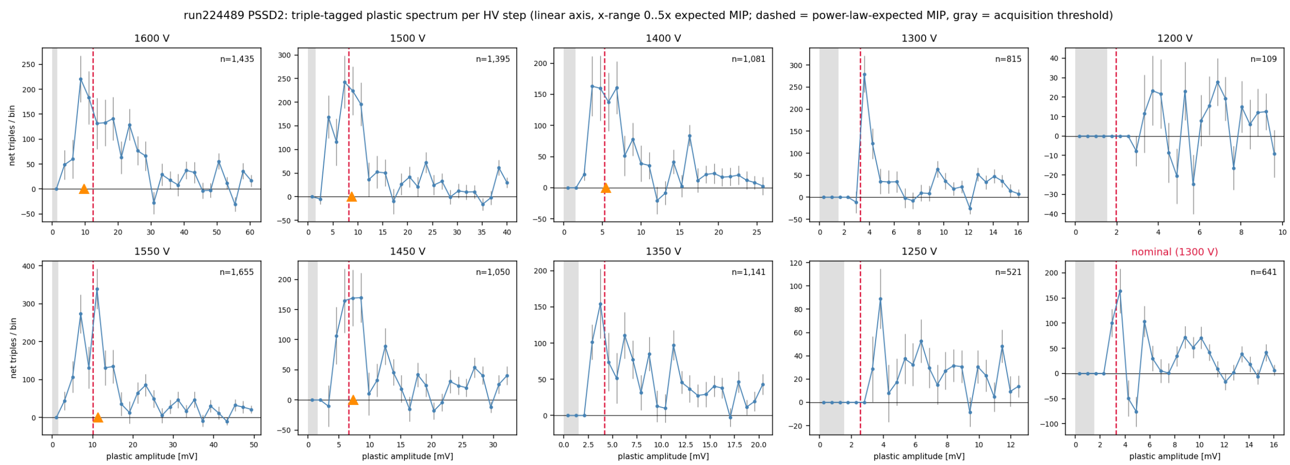

Figure 15: Same for PSSD2, the lowest-gain PMT: the peak is clear at $V \geq 1400\,\mathrm{V}$ but rides the threshold at low HV and at nominal — which is why only $V \geq 1400\,\mathrm{V}$ enters the fit, and why this arm cannot trigger on plastic MIPs at the current HV.

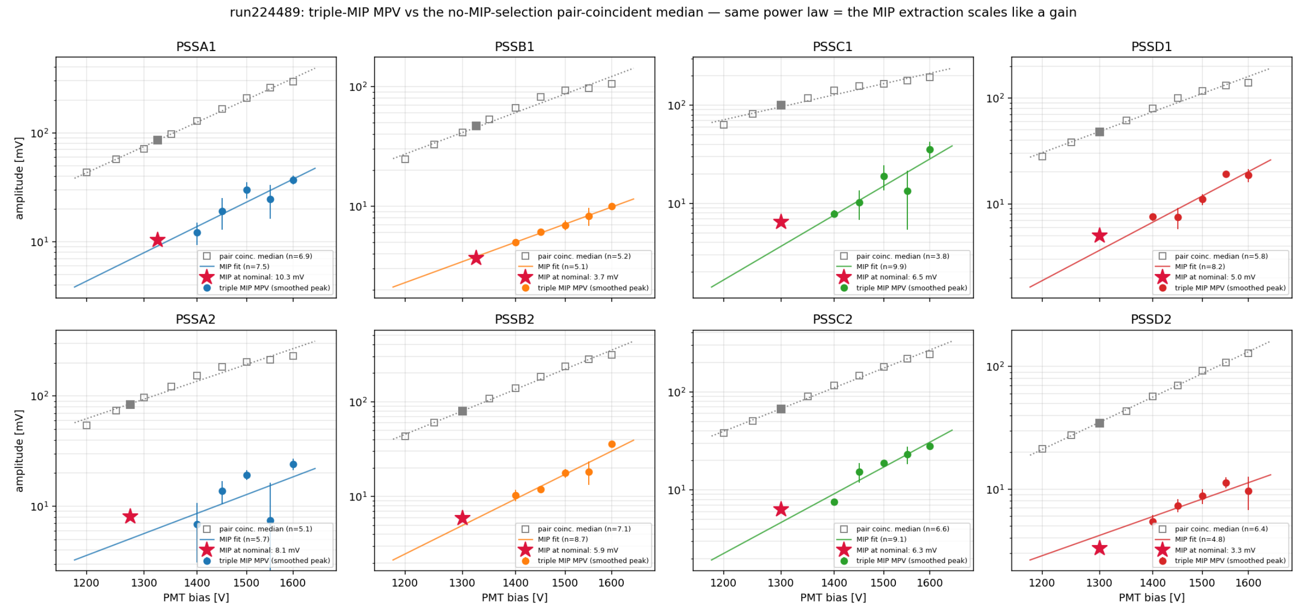

Figure 16: Per-step triple-MIP MPVs (colored, bootstrap errors, own power-law fit) vs. the pair-coincident median without any MIP selection (gray squares, dotted fit; filled square = nominal-HV window). Star: the adopted aligned-sum MPV at nominal HV. Same slope = the MIP extraction scales like a gain.

PMT	V_{nom} [V]	n	MPV [mV]	spread	mV/MeV
PSSA1 (AL)	1325	6.94	10.3 ± 1.5	16%	2.05
PSSA2 (AR)	1275	5.08	8.1 ± 1.2	42%	1.60
PSSB1 (BL)	1325	5.19	3.7 ± 0.5	2%	0.73
PSSB2 (BR)	1300	7.10	5.9 ± 0.5	16%	1.18
PSSC1 (CL)	1300	3.80	6.5 ± 1.6	38%	1.28
PSSC2 (CR)	1300	6.64	6.3 ± 0.3	18%	1.25
PSSD1 (DL)	1300	5.77	5.0 ± 0.6	18%	0.99
PSSD2 (DR)	1300	6.39	3.3 ± 0.3	14%	0.65

Table 3: Plastic MIP (linear-space MPV) at nominal HV and absolute calibration. MPV from the gain-aligned summed spectrum with bootstrap statistical error; "spread" is the per-step transport scatter (the fuller uncertainty on the weaker PMTs). The energy scale assumes the normal-incidence minimum-ionizing deposit in 2.5 cm of PVT ($1.956\,\mathrm{MeV\,cm^2/g} \times 1.032\,\mathrm{g/cm^3} \times 2.5\,\mathrm{cm} = 5.05\,\mathrm{MeV}$); the incidence-angle path-length spread ($\sim$+10–20% on the mean) is not yet corrected. `calib/pss_mip_calib_run224489.json` (the earlier log-binned-mode values are preserved there as `mip_mv_logmode`).

Two operational consequences. First, the MIP-based inter-PMT gain ratios agree with the coincident-median equalization to ∼15%, an independent validation of Table 2. Second, and important for the trigger: **at the current HV the DR (3.3 mV) and BL (3.7 mV) MIPs are below the 4.9 mV threshold** that the threshold scan recommends — those arms cannot trigger efficiently on plastic MIPs until the equalization (+133/+84 V) is applied.

6 LIQ gain vs. position: the gradient is real

The triple sample also locates each through-going track: horizontally by wall group (4 bins) and plastic bar (2), vertically by the top/bottom SiPM asymmetry $\ln(A_{\mathrm{top}}/A_{\mathrm{bot}})$ of the tagged wall hit (6 bins). Accumulating the LIQ amplitude in these bins (all double sideband-subtracted) gives Fig. 17. The marginal profiles are unambiguous:

- **A and D** (vessels HORIZONTAL, PMT toward wall group 4): the horizontal profile rises from 9.1→17.5 mV (A) and 6.4→11.1 mV (D) toward group 4, while the vertical profile is flat.
- **B and C** (vessels VERTICAL, PMT up): the vertical profile rises toward the top (11.7→17.5 mV on B, 4.5→7.8 mV on C), while the horizontal profile is mild.

A 1.5–2× light-collection gradient along the vessel axis, higher near the PMT, whose direction flips exactly with the surveyed vessel orientation (vertical PMT-up on B/C, horizontal PMT-along-$+u$ on A/D in the as-built Geant geometry). Any LIQ energy interpretation needs a position correction, which the triple machinery can now supply. The LIQ pulse-height scale itself is uncomfortably low on C and D (inclusive medians 4.7 and 5.3 mV vs. 10.4 mV on B): LIQ HV/gain review is advisable alongside the plastic equalization (Fig. 18).

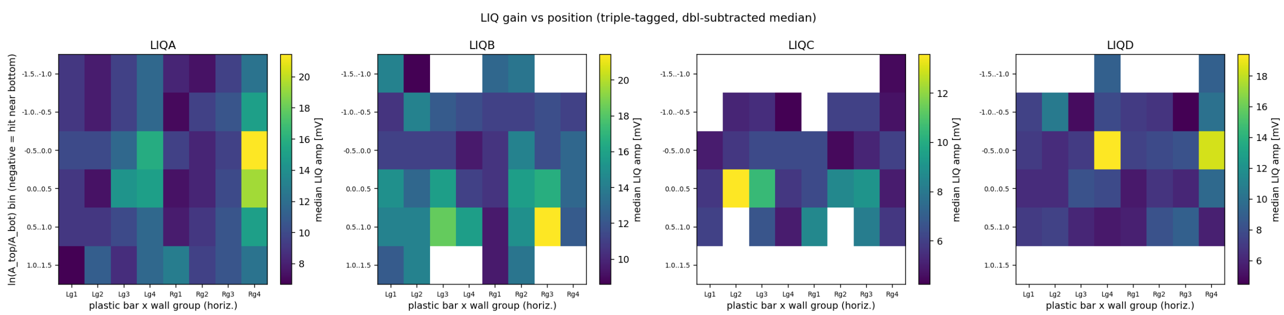

Figure 17: Median LIQ amplitude per position bin (triple-tagged, double sideband-subtracted). Columns: plastic bar × wall group (left→right as seen from the back); rows: vertical asymmetry bins (negative = hit near the bottom).

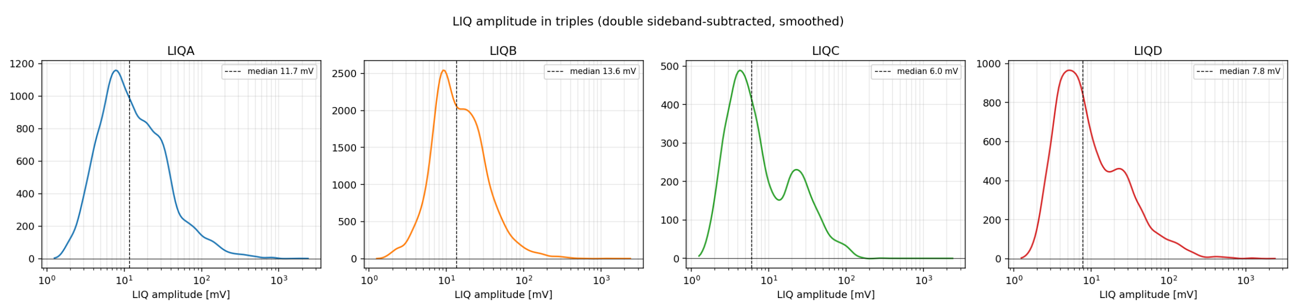

Figure 18: Triple-tagged LIQ amplitude spectra per arm.

7 Timing program

Common offset shift. The new per-channel wall–plastic offsets (`calib/time_offsets_run224489.json`) differ from the 224466 ones by a single common -22.27 ns with 0.22 ns rms across all 64 channel pairs (Fig. 19) — the FIFO path is a clean common delay (somewhat larger than the $\geq$16 ns cable estimate), and the sign matches "plastics later $\Rightarrow dt = t_{\text{wall}} - t_{\text{pss}}$ more negative".

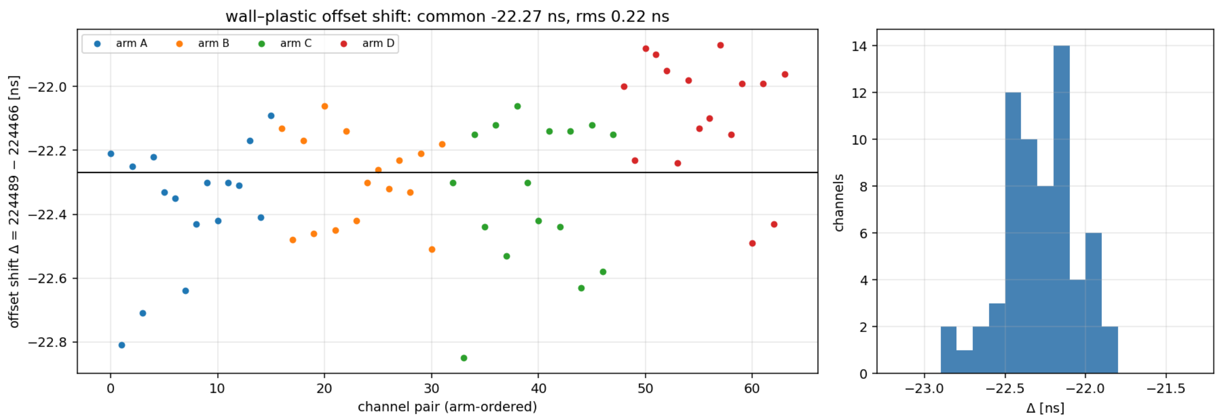

Figure 19: Per-channel offset shift between runs.

The -60 ns satellite is not a plastic-cable reflection. In 224466 the late-tof B/C dt spectra showed a small satellite at $dt \approx -60$ ns (6–9% of the main peak). If it were a reflection or afterpulse tied to the plastic signal path, the -22 ns path change would move it with the main peak. Instead it kept its *absolute* position (-60 to -68 ns) while the main peak moved by -22 ns — and it grew to 19–34% of the main peak (Fig. 20). Both facts point at something injected at or after the FIFO (digitizer side), and the growth suggests the FIFO itself participates. It stays outside all signal and sideband windows, so no result here is affected, but a scope check at the FIFO outputs is recommended.

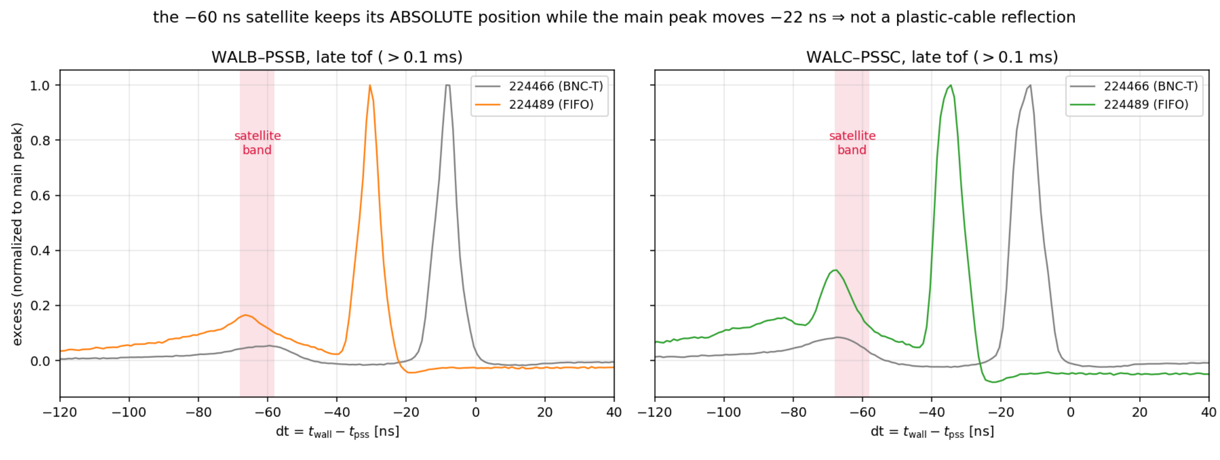

Figure 20: Late-tof dt spectra, normalized to the main peak, both runs.

8 Summary and recommendations

1. The A/D cross-wiring is fixed and fully explains the old A/D coincidence deficit; the wall MIP calibration now covers all four arms.

2. The plastic MIP is measured for the first time: 0.65–2.05 mV/MeV at nominal HV. **Apply the new equalization** (Table 2); until then DR and BL cannot trigger on plastic MIPs at the recommended 4.9 mV threshold.
3. Use the measured per-PMT FIFO ratios (×1.13–1.65), not a global ×2, in any comparison with pre-FIFO data.
4. The liquids are timed in (offsets published); the LIQ position-gain gradient (1.5–2×, toward the PMT) must be corrected before any LIQ energy use; C/D LIQ gains look low — review LIQ HV.
5. Scope-check the FIFO outputs for the −60 ns satellite, which tripled in relative size with the new path.

Reproducibility: caches and calibrations are keyed by run stem under `mx_july_beam_qa/` (`cache/*_run224489.npz`, `calib/*_run224489.json`); triples machinery in `19_liq_triples.py` (+`19b`, `19c`); HV-scan step tables per run in `12_plastic_hv_scan.py`.